

HiSSE

Hidden State Speciation and Extinction

Modelos de extinción y especiación dependientes de estados ocultos

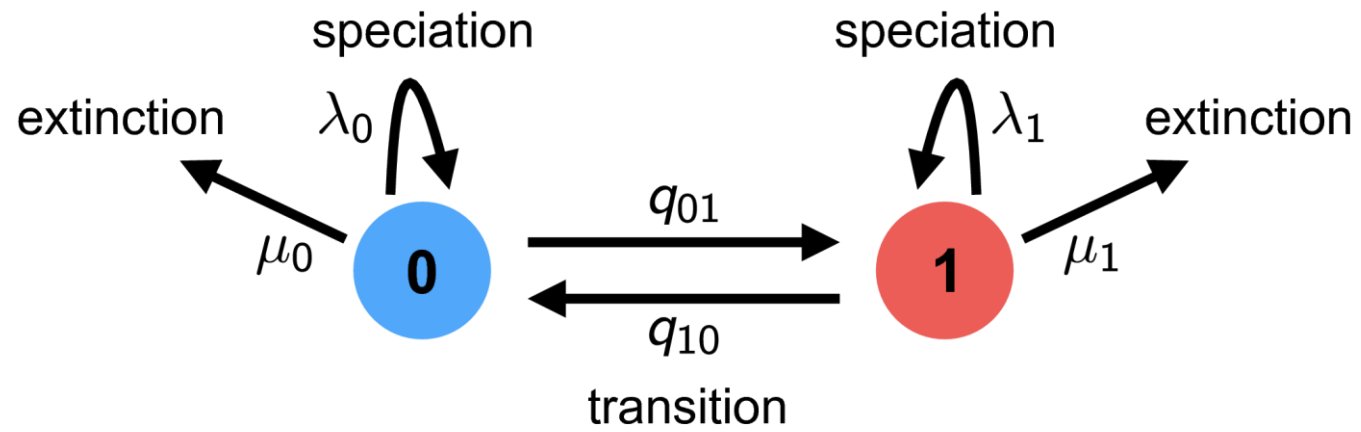
Contenido

1. Recordatorio : BiSSE y otros modelos SSE.
 - BiSSE
 - Un ejemplo particular: Multitrait MuSSE
2. Críticas de BiSSE
 - Maddison & FitzJohn, 2015
 - Rabosky & Goldberg, 2014
3. HiSSE como una solución.
 - Beaulieu & O'Meara, 2016.

1) Recordatorio: Modelos SSE

BiSSE

BiSSE

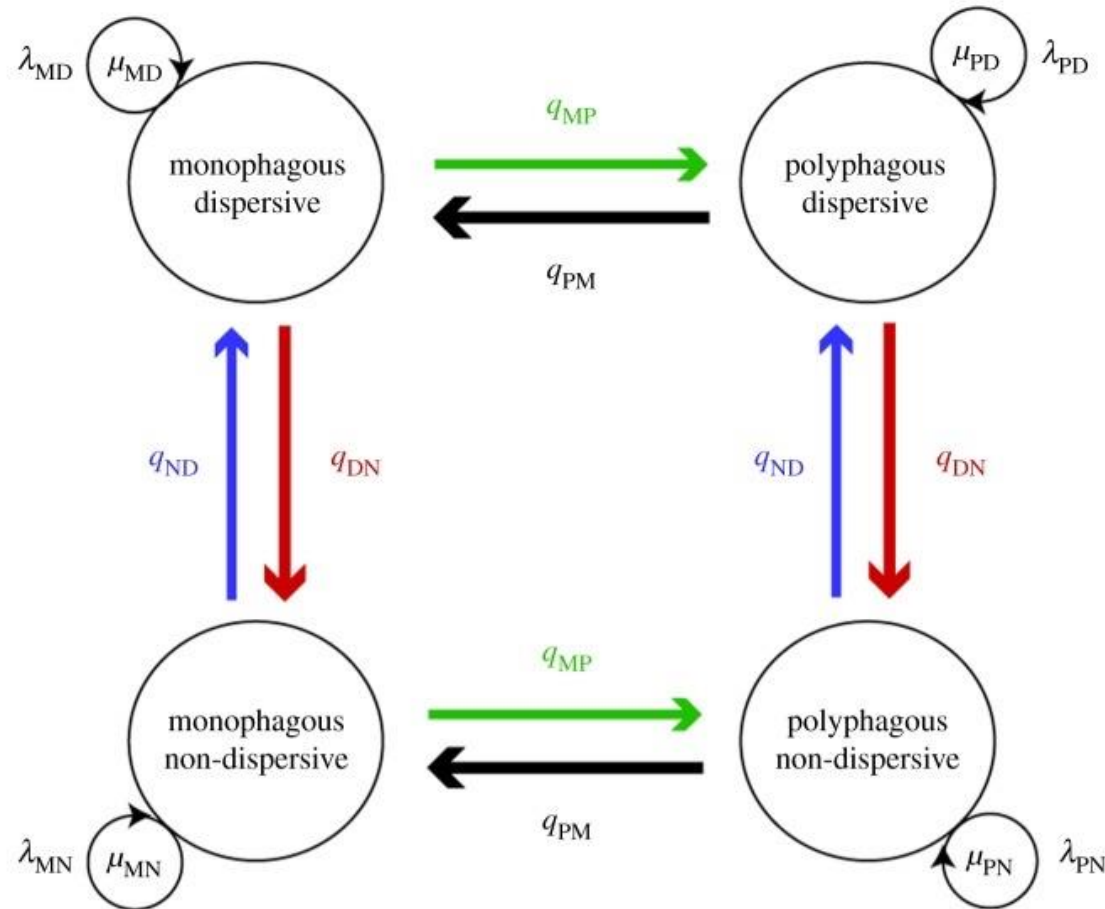


MuSSE (con cuatro estados de caracter)

Multi-trait MuSSE

Multi-trait MuSSE

(a)



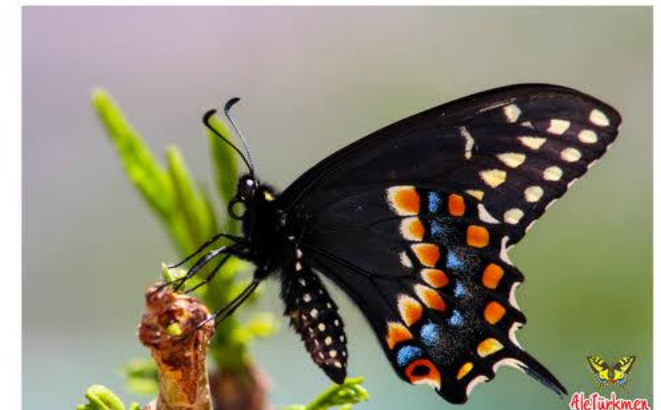
► *Proc Biol Sci.* 2014 Nov 22;281(1795):20132960. doi: [10.1098/rspb.2013.2960](https://doi.org/10.1098/rspb.2013.2960)

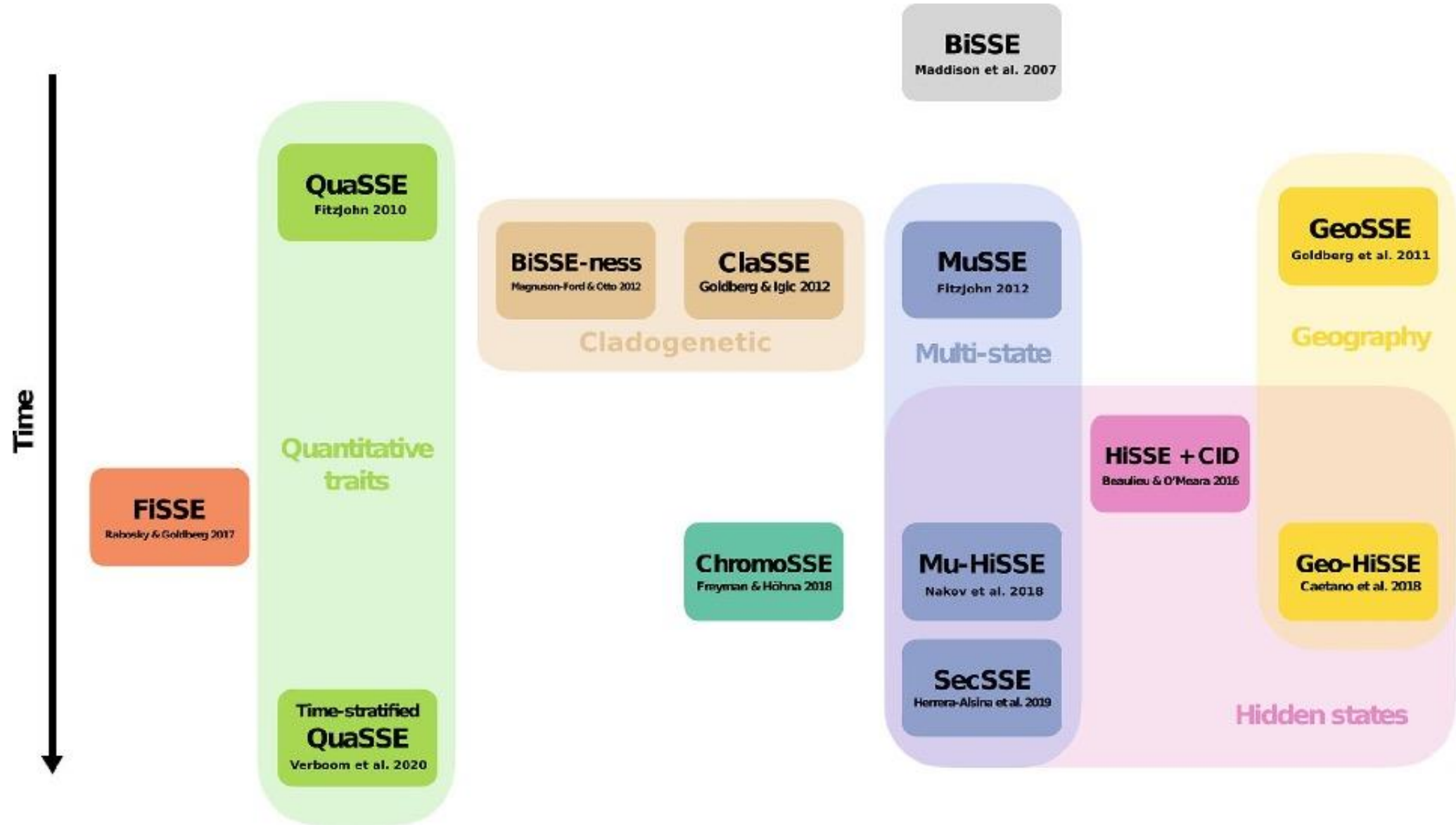
Specialization and generalization in the diversification of phytophagous insects: tests of the musical chairs and oscillation hypotheses

[Nate B Hardy](#) ^{1,*}, [Sarah P Otto](#) ^{2,*}

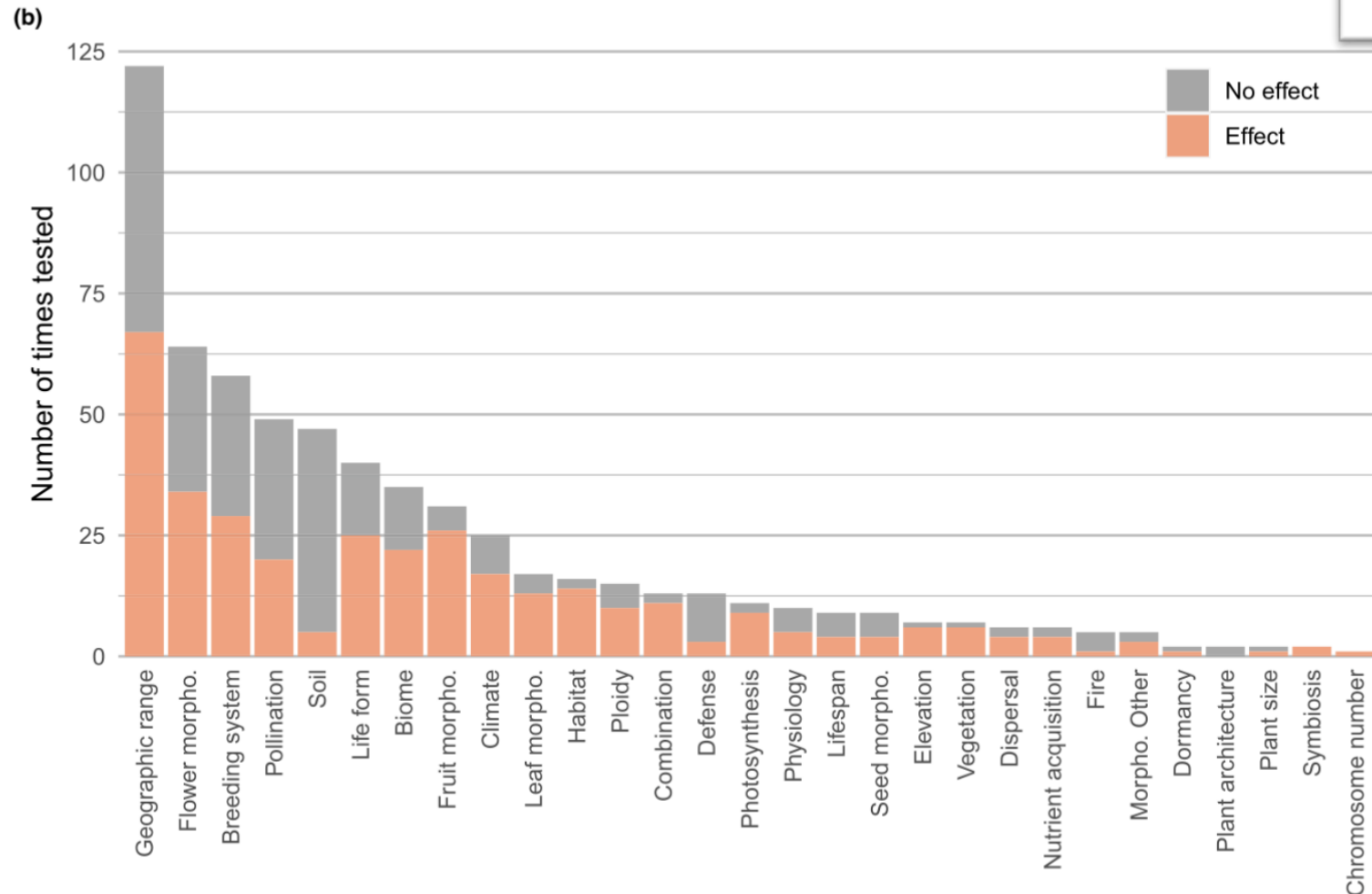
► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC4213601 PMID: [25274368](https://pubmed.ncbi.nlm.nih.gov/25274368/)





La popularidad de los modelos SSE



Received: 18 May 2022 | Revised: 19 December 2022 | Accepted: 22 December 2022

DOI: 10.1111/ele.14179

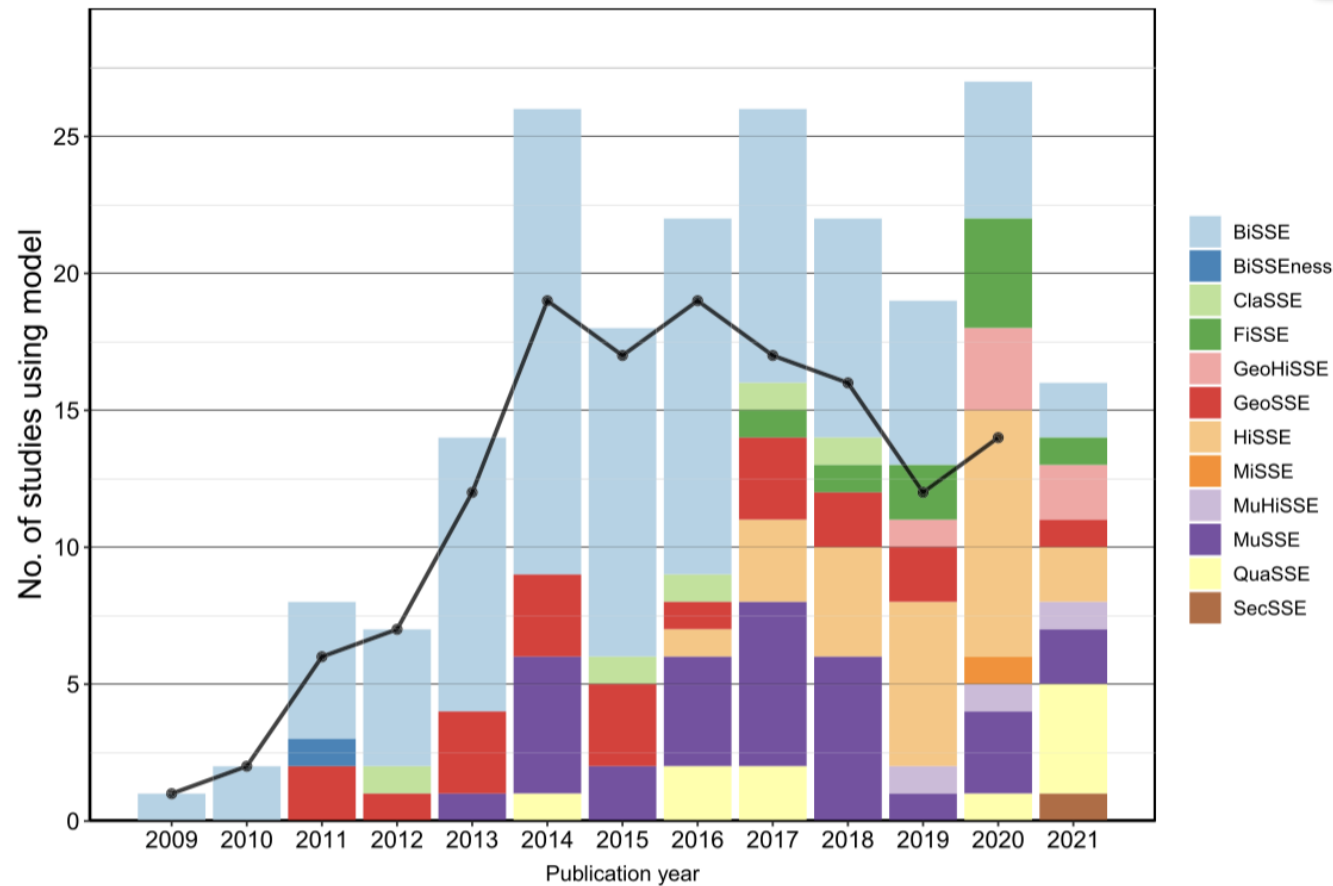
SYNTHESIS

ECOLOGY LETTERS WILEY

Trait-dependent diversification in angiosperms: Patterns, models and data

Andrew J. Helmstetter¹ | Rosana Zenil-Ferguson² | Hervé Sauquet^{3,4} | Sarah P. Otto⁵ | Marcos Méndez⁶ | Mario Vallejo-Marin⁷ | Jürg Schönenberger⁸ | Concetta Burgarella⁹ | Bruce Anderson¹⁰ | Hugo de Boer¹¹ | Sylvain Glémin^{7,12} | Jos Käfer^{13,14}

La popularidad de los modelos SSE



Received: 18 May 2022 | Revised: 19 December 2022 | Accepted: 22 December 2022

DOI: 10.1111/ele.14179

SYNTHESIS

ECOLOGY LETTERS WILEY

Trait-dependent diversification in angiosperms: Patterns, models and data

Andrew J. Helmstetter¹ | Rosana Zenil-Ferguson² | Hervé Sauquet^{3,4} | Sarah P. Otto⁵ | Marcos Méndez⁶ | Mario Vallejo-Marin⁷ | Jürg Schönenberger⁸ | Concetta Burgarella⁹ | Bruce Anderson¹⁰ | Hugo de Boer¹¹ | Sylvain Glémin^{7,12} | Jos Käfer^{13,14}

2) Problemas con BiSSE

Crítica 1 (Maddison & FitzJohn, 2015)

JOURNAL ARTICLE

The Unsolved Challenge to Phylogenetic Correlation Tests for Categorical Characters

Wayne P. Maddison , Richard G. FitzJohn [Author Notes](#)

Systematic Biology, Volume 64, Issue 1, January 2015, Pages 127–136,

<https://doi.org/10.1093/sysbio/syu070>

Published: 09 September 2014 [Article history](#) ▼

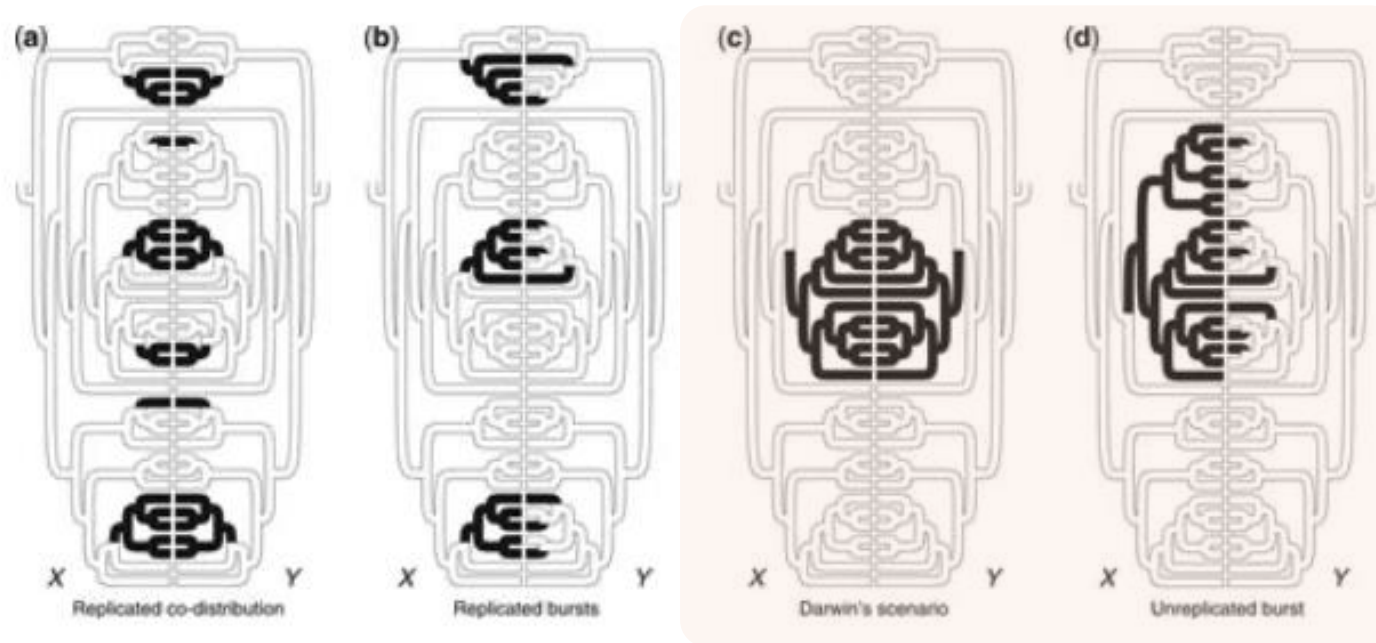


FIGURE 1. Four scenarios for the evolution of states of characters X and Y . In each, the same phylogeny is mirrored to show X at left and Y at right. State 0=white; State 1=black. (a) Replicated co-distribution. (b) Replicated bursts. (c) Darwin's scenario. (d) Unreplicated burst. Panels (a) and (b) provide good evidence for an interesting adaptive/functional relationship between X and Y ; panels (c) and (d) do not.

Crítica 1 (Maddison & FitzJohn, 2015)

JOURNAL ARTICLE

The Unsolved Challenge to Phylogenetic Correlation Tests for Categorical Characters

Wayne P. Maddison , Richard G. FitzJohn  [Author Notes](#)

Systematic Biology, Volume 64, Issue 1, January 2015, Pages 127–136,

<https://doi.org/10.1093/sysbio/syu070>

Published: 09 September 2014 [Article history](#) ▼

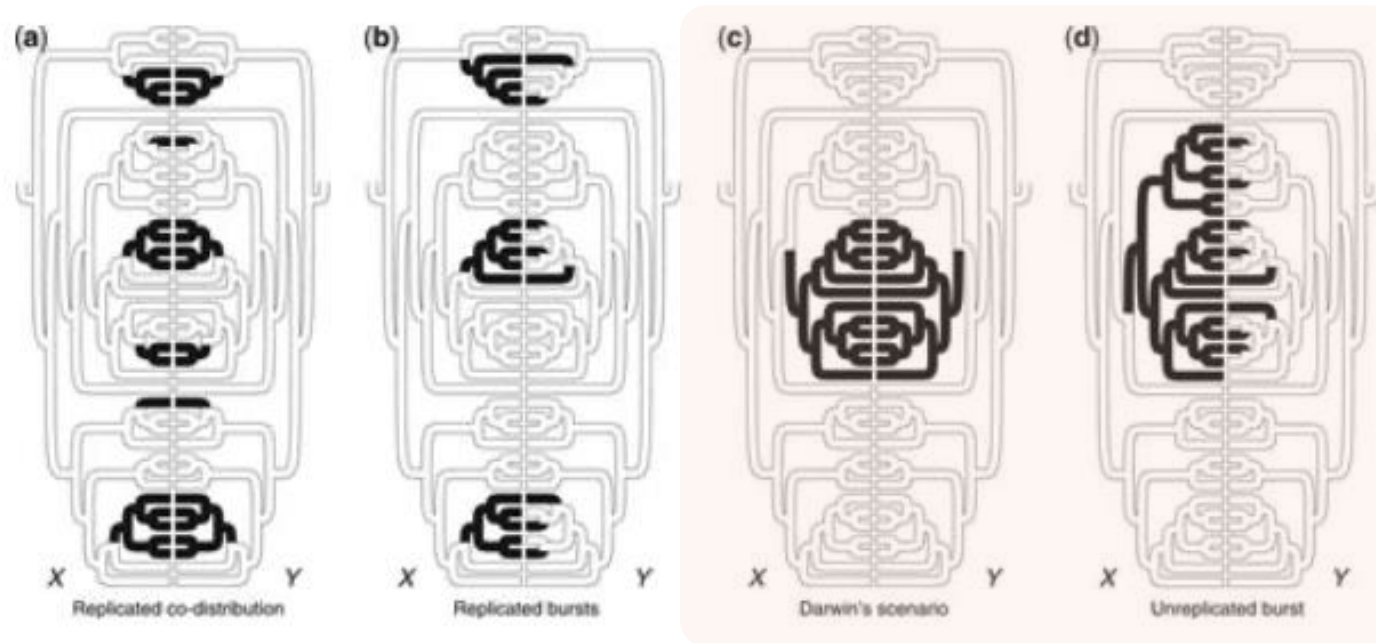
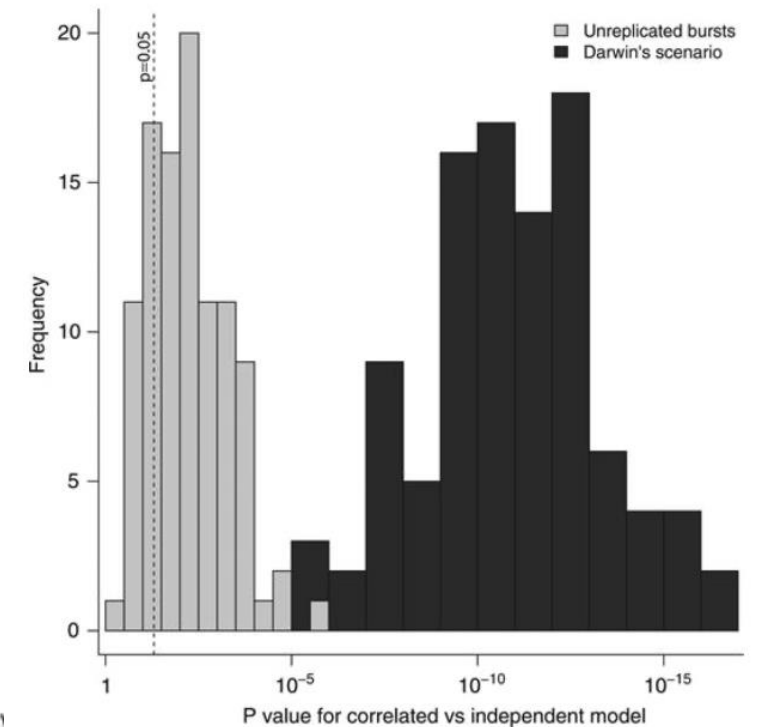


FIGURE 1. Four scenarios for the evolution of states of characters X and Y . In each, the same phylogeny is mirrored to show X at left and Y at right. State 0=white; State 1=black. (a) Replicated co-distribution. (b) Replicated bursts. (c) Darwin's scenario. (d) Unreplicated burst. Panels (a) and (b) provide good evidence for an interesting adaptive/functional relationship between X and Y ; panels (c) and (d) do not.



Crítica 2 (Rabosky & Goldberg, 2014)

Model Inadequacy and Mistaken Inferences of Trait-Dependent Speciation

DANIEL L. RABOSKY^{1,*} AND EMMA E. GOLDBERG²

¹Museum of Zoology and Department of Ecology and Evolutionary Biology, University of Michigan, Ann Arbor, MI 48109; and ²Department of Ecology, Evolution, and Behavior, University of Minnesota, Saint Paul, MN 55108, USA

*Correspondence to be sent to: Museum of Zoology and Department of Ecology and Evolutionary Biology, University of Michigan, Ann Arbor, MI 48109 USA; E-mail: drabosky@umich.edu.

Received 18 June 2014; reviews returned 21 December 2014; accepted 22 December 2014

Associate Editor: Vincent Savolainen

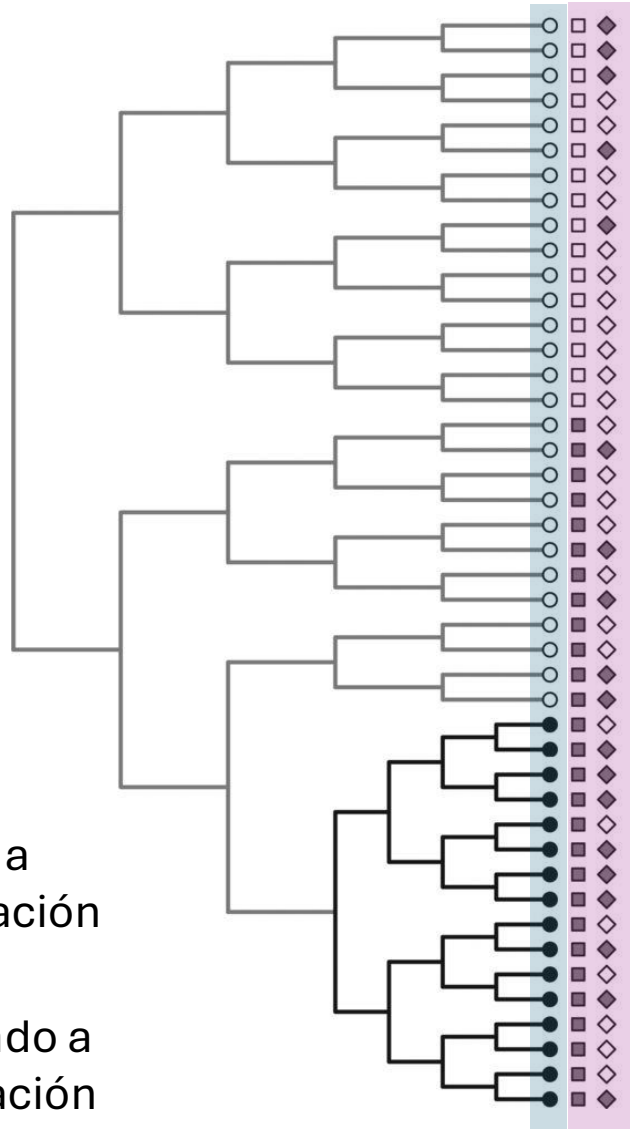
Daniel L. Rabosky and Emma E. Goldberg contributed equally to this article.

Abstract.—Species richness varies widely across the tree of life, and there is great interest in identifying ecological, geographic, and other factors that affect rates of species proliferation. Recent methods for explicitly modeling the relationships among character states, speciation rates, and extinction rates on phylogenetic trees—BiSSE, QuaSSE, GeoSSE, and related models—have been widely used to test hypotheses about character state-dependent diversification rates. Here, we document the disconcerting ease with which neutral traits are inferred to have statistically significant associations with speciation rate. We first demonstrate this unfortunate effect for a known model assumption violation: shifts in speciation rate associated with a character not included in the model. We further show that **for many empirical phylogenies, characters simulated in the absence of state-dependent diversification exhibit an even higher Type I error rate, indicating that the method is susceptible to additional, unknown model inadequacies.** For traits that evolve slowly, the root cause appears to be a statistical framework that does not require replicated shifts in character state and diversification. However, spurious associations between character state and speciation rate arise even for traits that lack phylogenetic signal, suggesting that phylogenetic pseudoreplication alone cannot fully explain the problem. **The surprising severity of this phenomenon suggests that many trait-diversification relationships reported in the literature may not be real.** More generally, we highlight the need for diagnosing and understanding the consequences of model inadequacy in phylogenetic comparative methods. [Character evolution; extinction; macroevolution; speciation; statistics.]

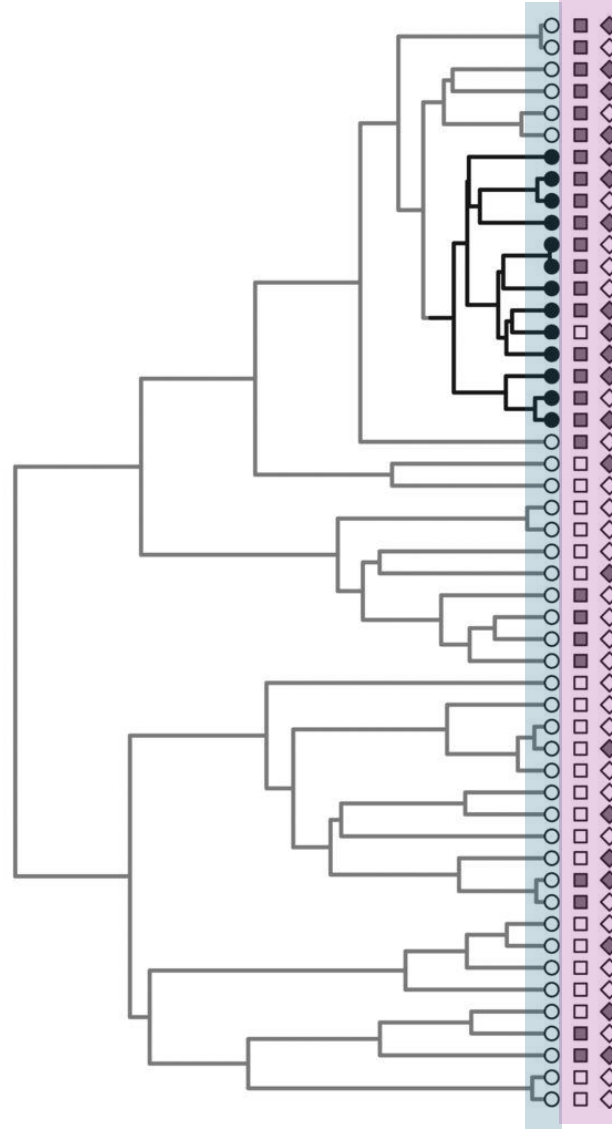
Crítica 2 (Rabosky & Goldberg, 2014)

SIMULACIONES

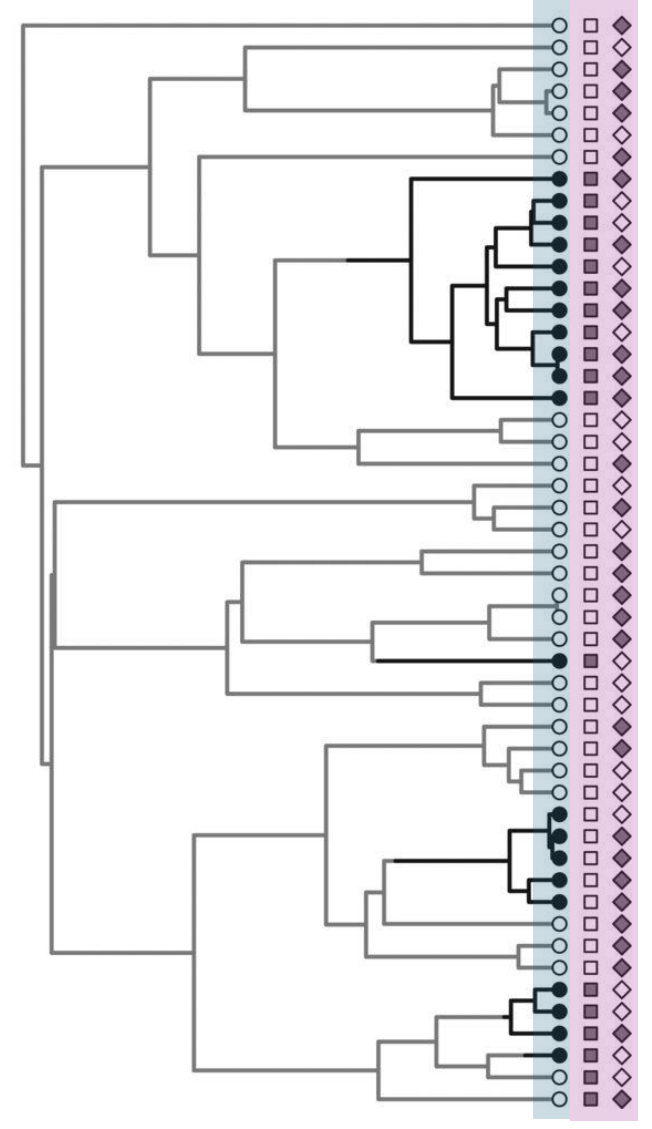
A) one diversification shift, idealized



B) rare diversification shifts, simulated



C) common diversification shifts, simulated

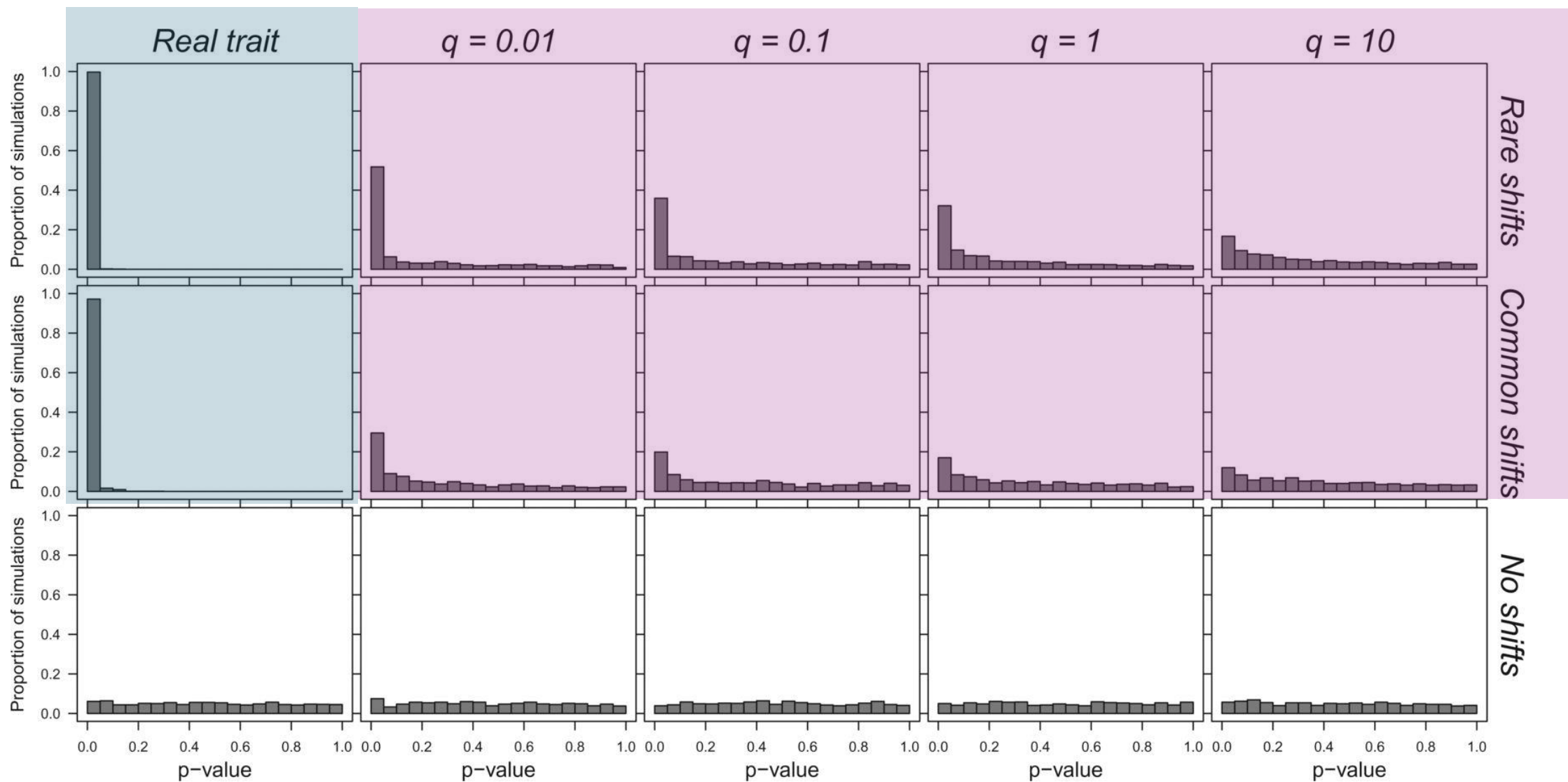


Asociado a diversificación

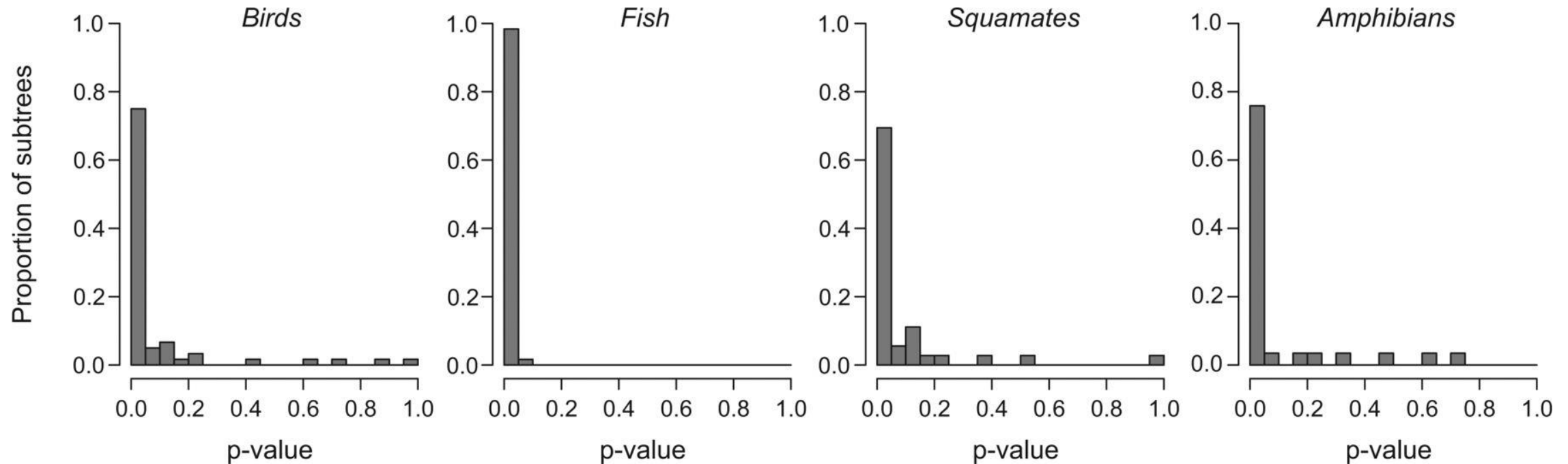
No asociado a diversificación

Crítica 2 (Rabosky & Goldberg, 2014)

INFERENCIA USANDO BISSE



Efecto de la longitud del nombre de las especies en las tasas de diversificación



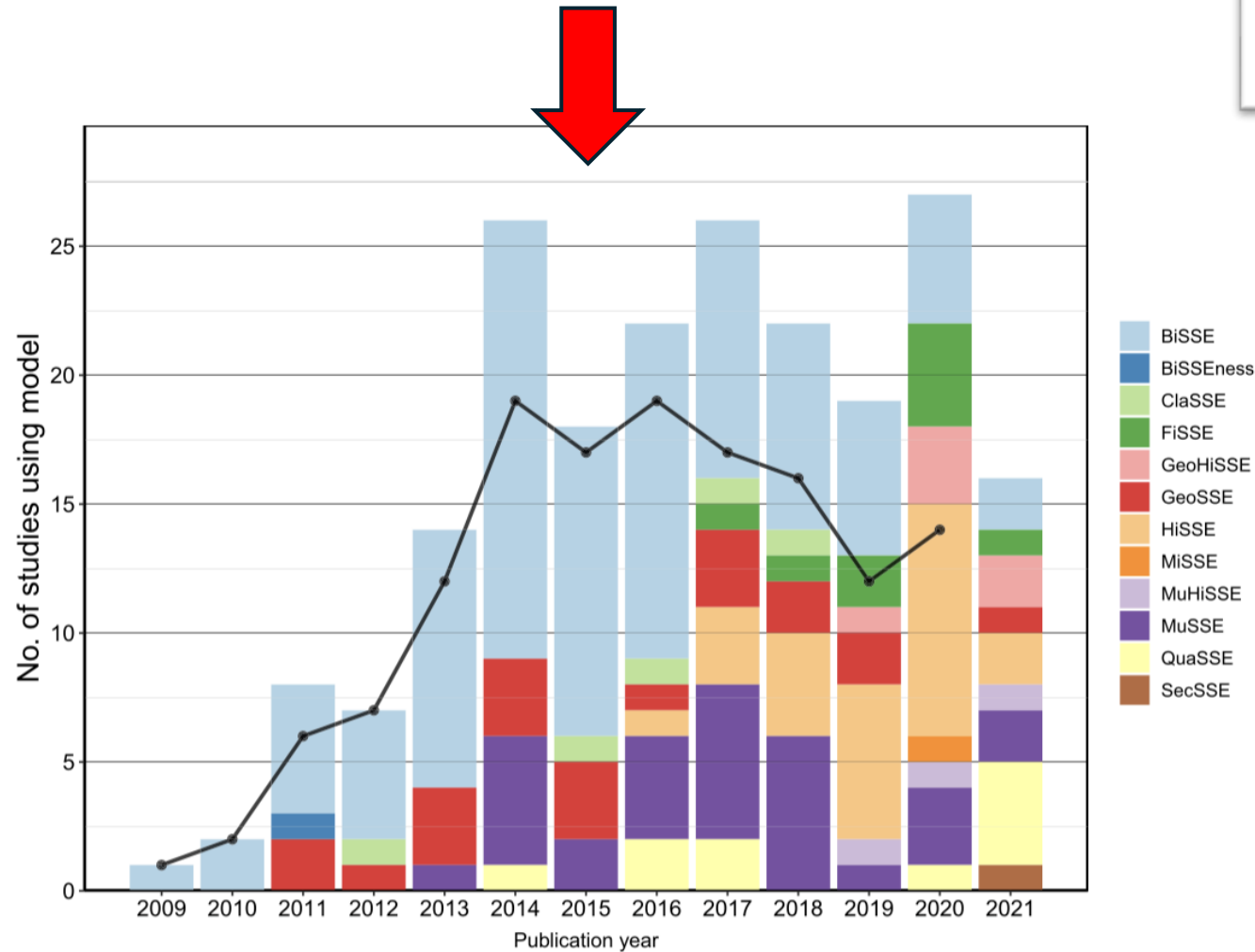
Distribution of values for tests of the effect of taxon name length on speciation rate for phylogenies of birds, fishes, squamates, and amphibians; value is the probability of the data under the null hypothesis that taxon name length is not associated with speciation rate. Taxon name length was scored as a binary character (short, long) depending on whether the number of letters in the Latin binomial exceeded the median name length in the tree. Overall, taxon name length is frequently associated with speciation rate.

BiSSE desde una perspectiva conceptual

- ¿Hay variación en el proceso de diversificación asociada al caracter focal?
 - Premisa 1: hay variación
 - Premisa 2: asociada al caracter focal

BiSSE no le da oportunidad a la variación en el proceso de diversificación de estar asociada con otro proceso, e.g. “variación de fondo”, otro(s) caracter(es).

La popularidad de los modelos SSE



Received: 18 May 2022 | Revised: 19 December 2022 | Accepted: 22 December 2022

DOI: 10.1111/ele.14179

SYNTHESIS

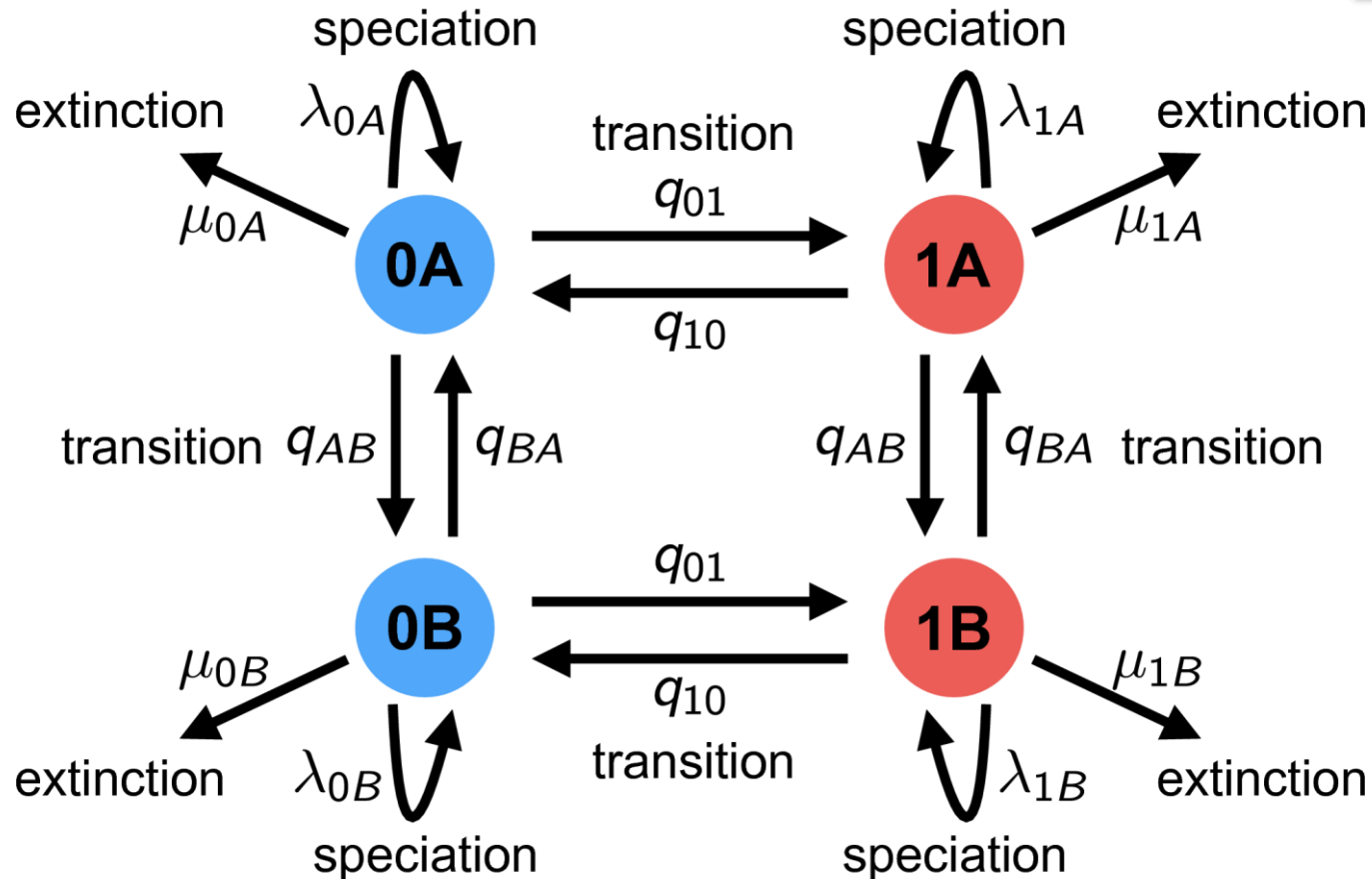
ECOLOGY LETTERS WILEY

Trait-dependent diversification in angiosperms: Patterns, models and data

Andrew J. Helmstetter¹ | Rosana Zenil-Ferguson² | Hervé Sauquet^{3,4} | Sarah P. Otto⁵ | Marcos Méndez⁶ | Mario Vallejo-Marin⁷ | Jürg Schönenberger⁸ | Concetta Burgarella⁹ | Bruce Anderson¹⁰ | Hugo de Boer¹¹ | Sylvain Glémin^{7,12} | Jos Käfer^{13,14}

3) HiSSE

HiSSE



Appl. Biol. 45(4):403-405, 2016
 © The Author(s) 2016. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved.
 For Permissions, please email: journals.permissions@oup.com
 DOI:10.1093/apbi/apw022
 Advance Access publication March 25, 2016

Detecting Hidden Diversification Shifts in Models of Trait-Dependent Speciation and Extinction

JEREMY M. BEAULIEU^{1,2,*} AND BRIAN C. O'MEARA²

¹National Institute for Biological and Mathematical Synthesis, University of Tennessee, Knoxville, TN 37996, USA and ²Department of Ecology and Evolutionary Biology, University of Tennessee, Knoxville, TN 37996-1610, USA

*Correspondence to be sent to: National Institute for Biological and Mathematical Synthesis, University of Tennessee, Knoxville, TN 37996, USA; Email: jbeaulieu@nimbios.org

Received 10 March 2015; reviews returned 8 March 2016; accepted 8 March 2016
 Associate Editor: Travis Crandall

- Un estado **observado**: el estado de interés.
- Un estado **oculto**: este estado es capaz de “absorber” cambios en el proceso de diversificación que no están relacionados con el estado observado.
- No tenemos información del estado oculto de los terminals.

HiSSE en acción

Syst. Biol. 68(3):505–519, 2019

© The Author(s) 2018. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved.

For permissions, please email: journals.permissions@oup.com

DOI:10.1093/sysbio/syy078

Advance Access publication November 26, 2018

Stochastic Character Mapping of State-Dependent Diversification Reveals the Tempo of Evolutionary Decline in Self-Compatible Onagraceae Lineages

WILLIAM A. FREYMAN^{1,*} AND SEBASTIAN HÖHNA²

¹Department of Integrative Biology, University of California, Berkeley, 3040 Valley Life Sciences Building #3140, CA 94720, USA; and ²Division of Evolutionary Biology, Ludwig-Maximilians-Universität München, Geschwister-Scholl-Platz 1, 80539 Munich, Germany

*Correspondence to be sent to: Department of Integrative Biology, University of California, Berkeley, CA 94720, USA;
E-mail: willfreyman@gmail.com.

Received 16 January 2018; reviews returned 05 November 2018; accepted 13 November 2018

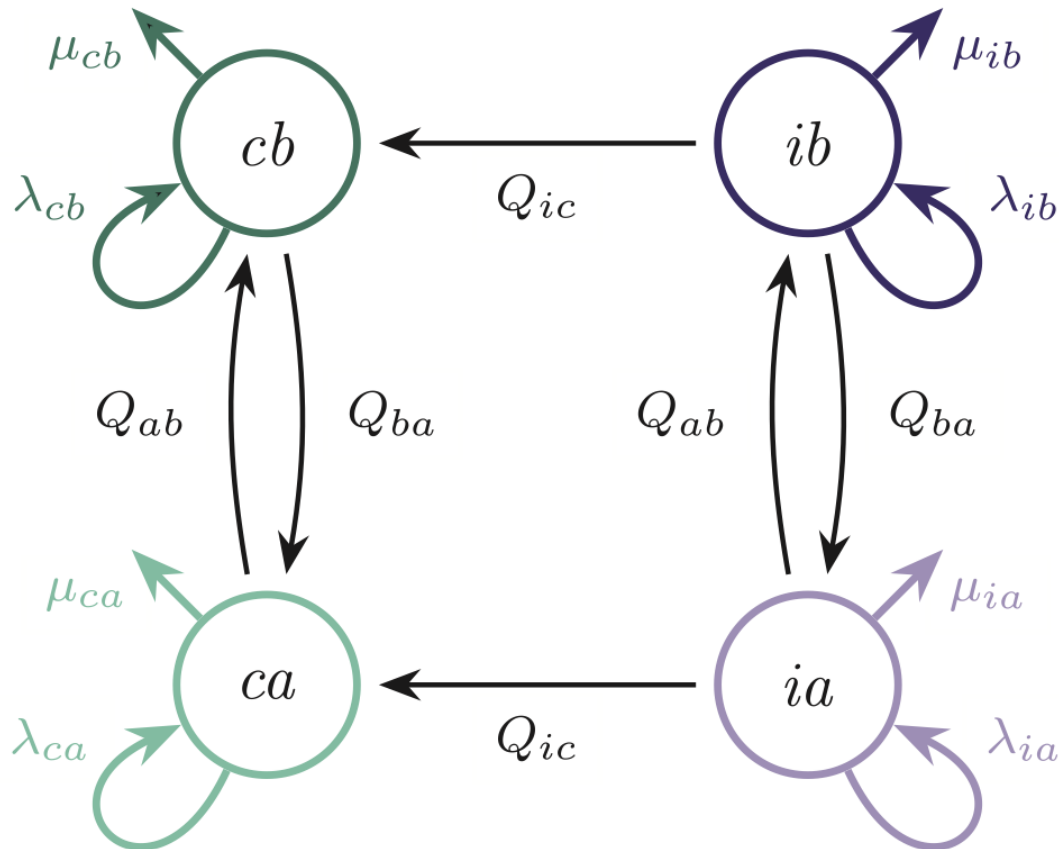
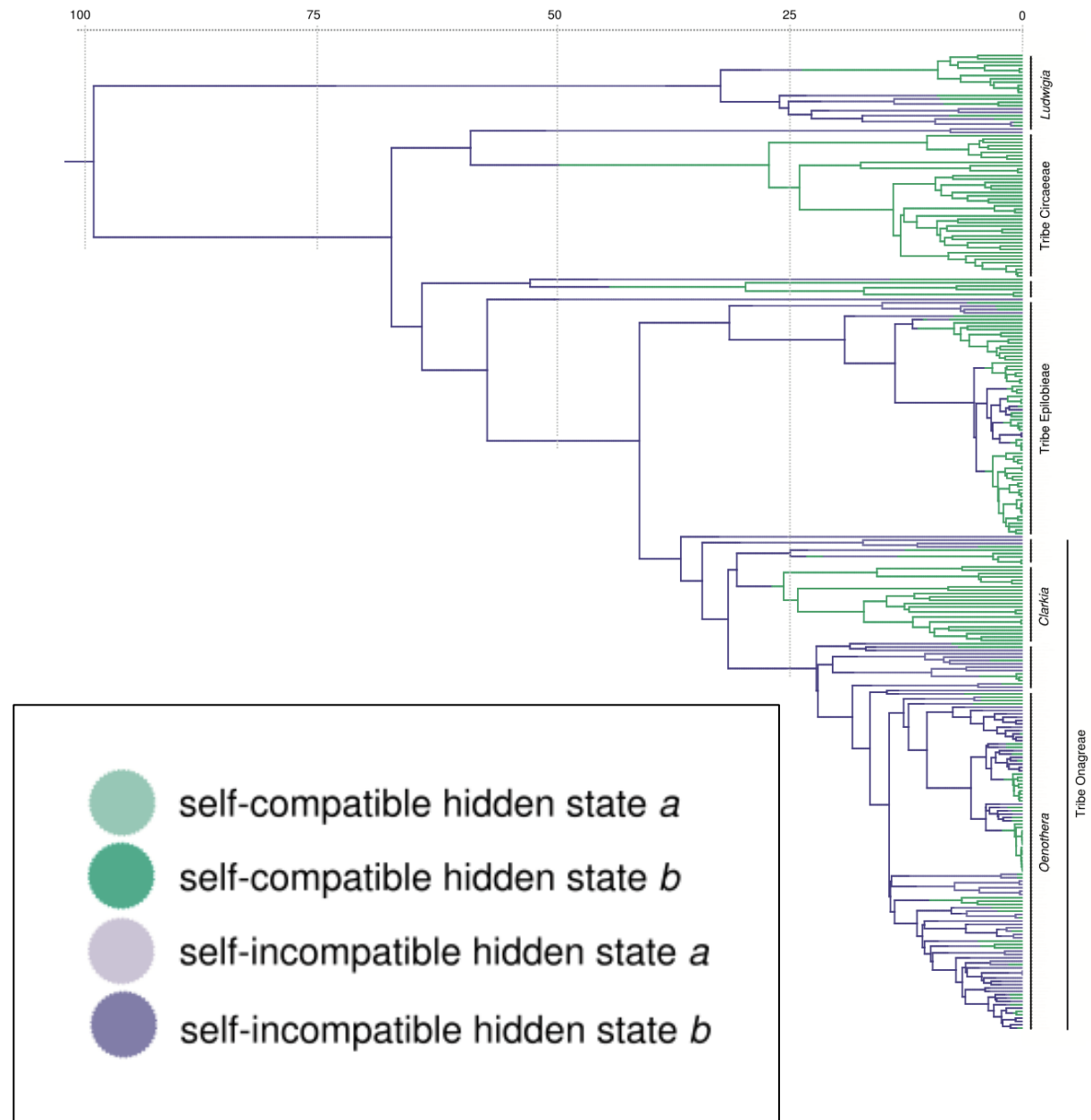
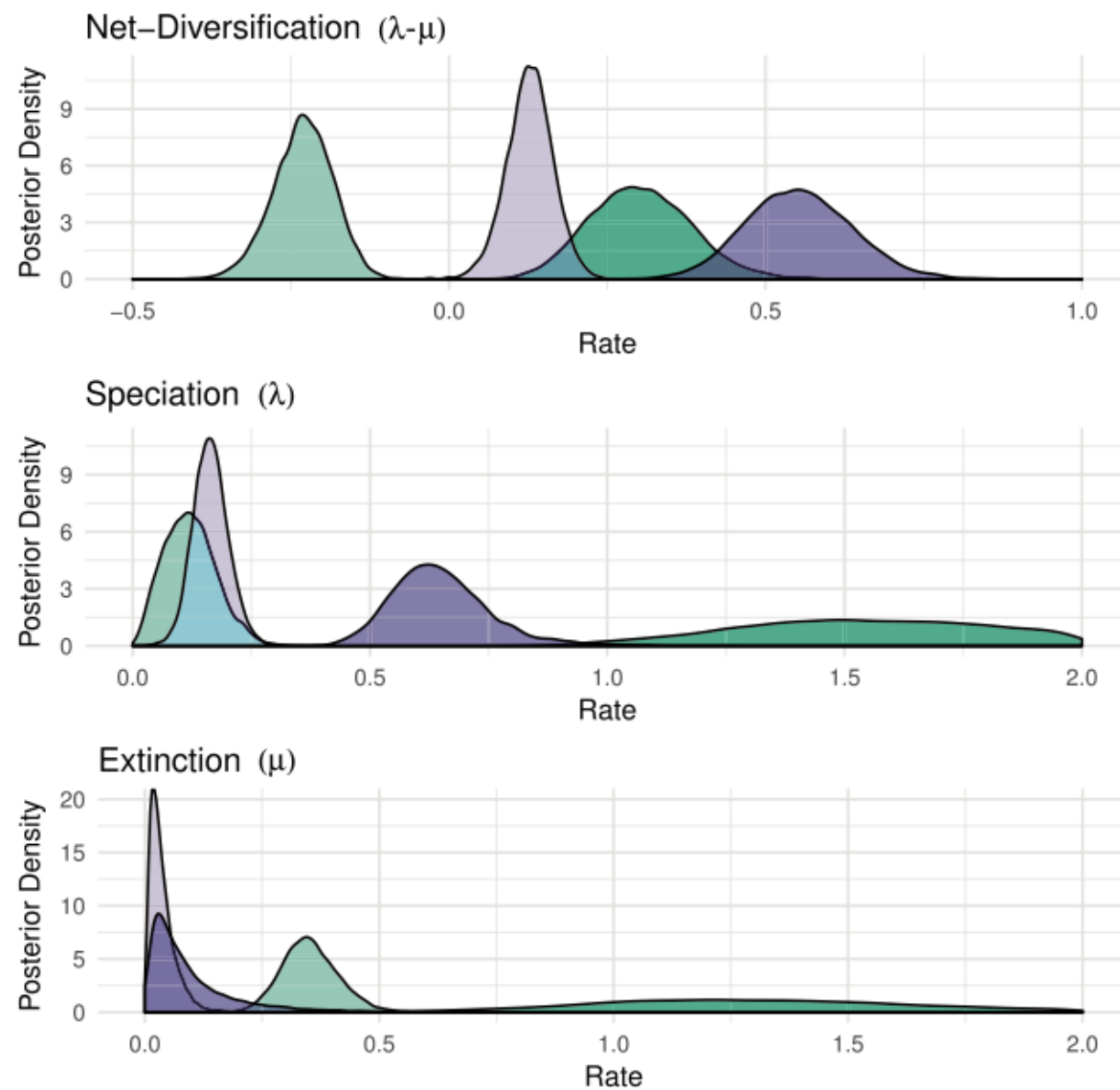


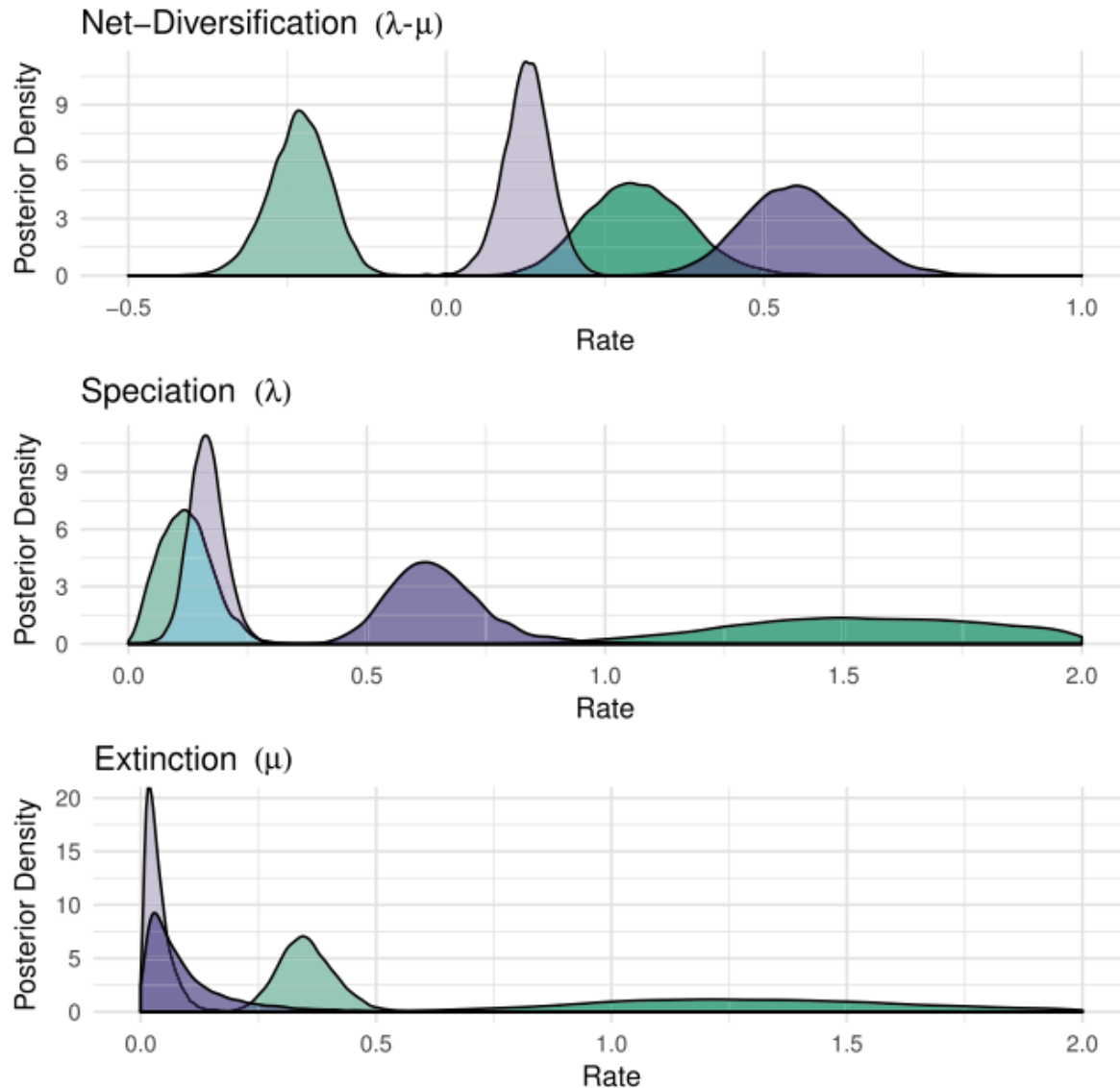
FIGURE 5. SSE model depicting states and rate parameters used to infer mating system evolution. The states are labeled *ca*, *cb*, *ia*, and *ib*, representing self-compatible hidden state a, self-compatible hidden state b, self-incompatible hidden state a, and self-incompatible hidden state b, respectively. Independent extinction and speciation rates were estimated for each of the four states, as well as the rate of transitioning from self-incompatible to self-compatible Q_{ic} and the rates of transitioning between the hidden states Q_{ab} and Q_{ba} .



HiSSE en acción

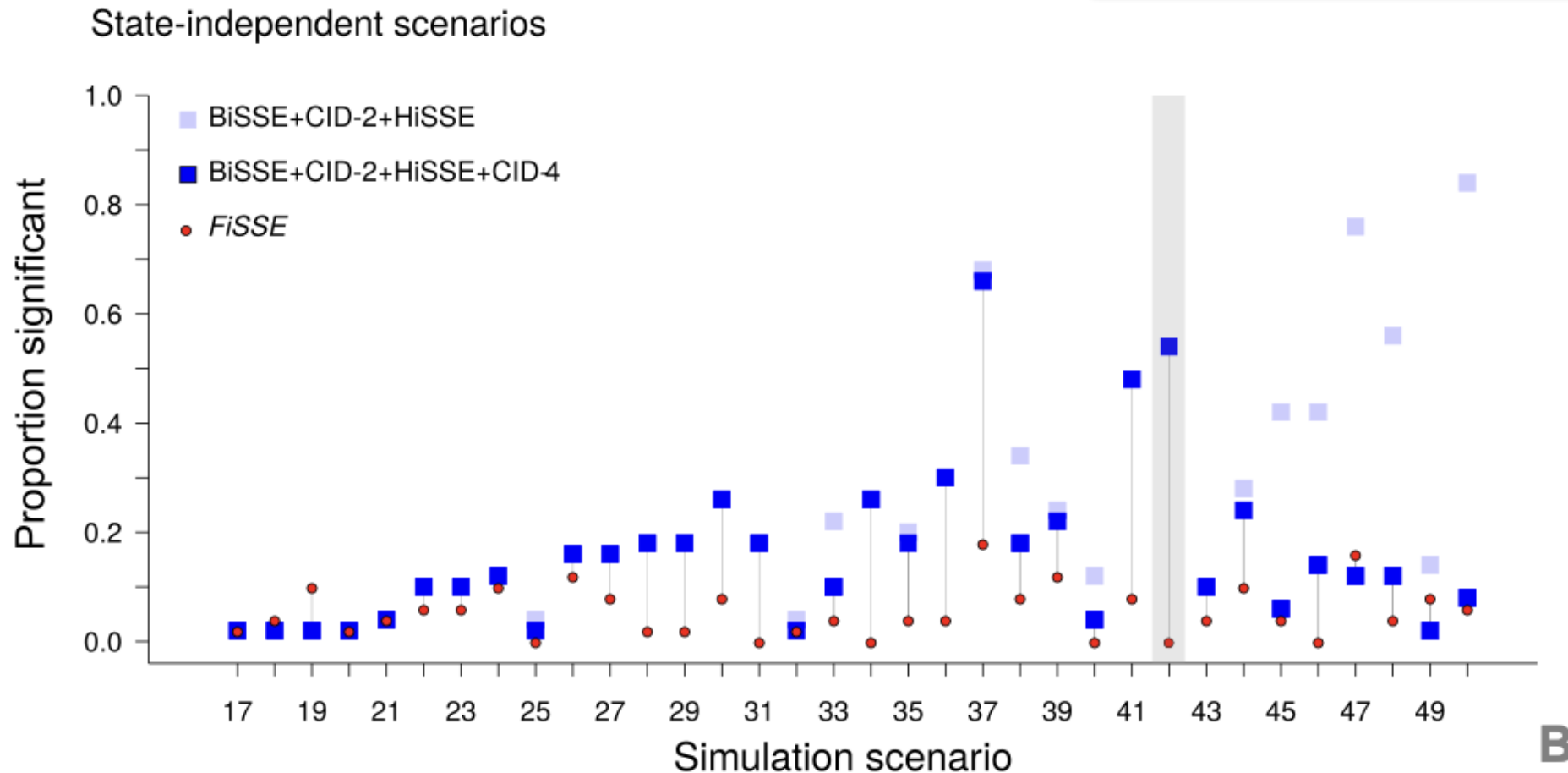
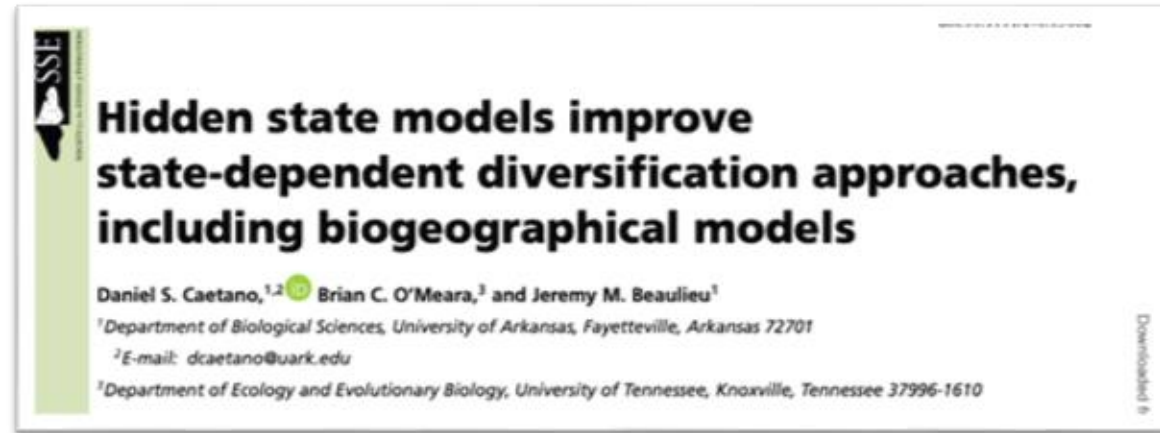


HiSSE en acción



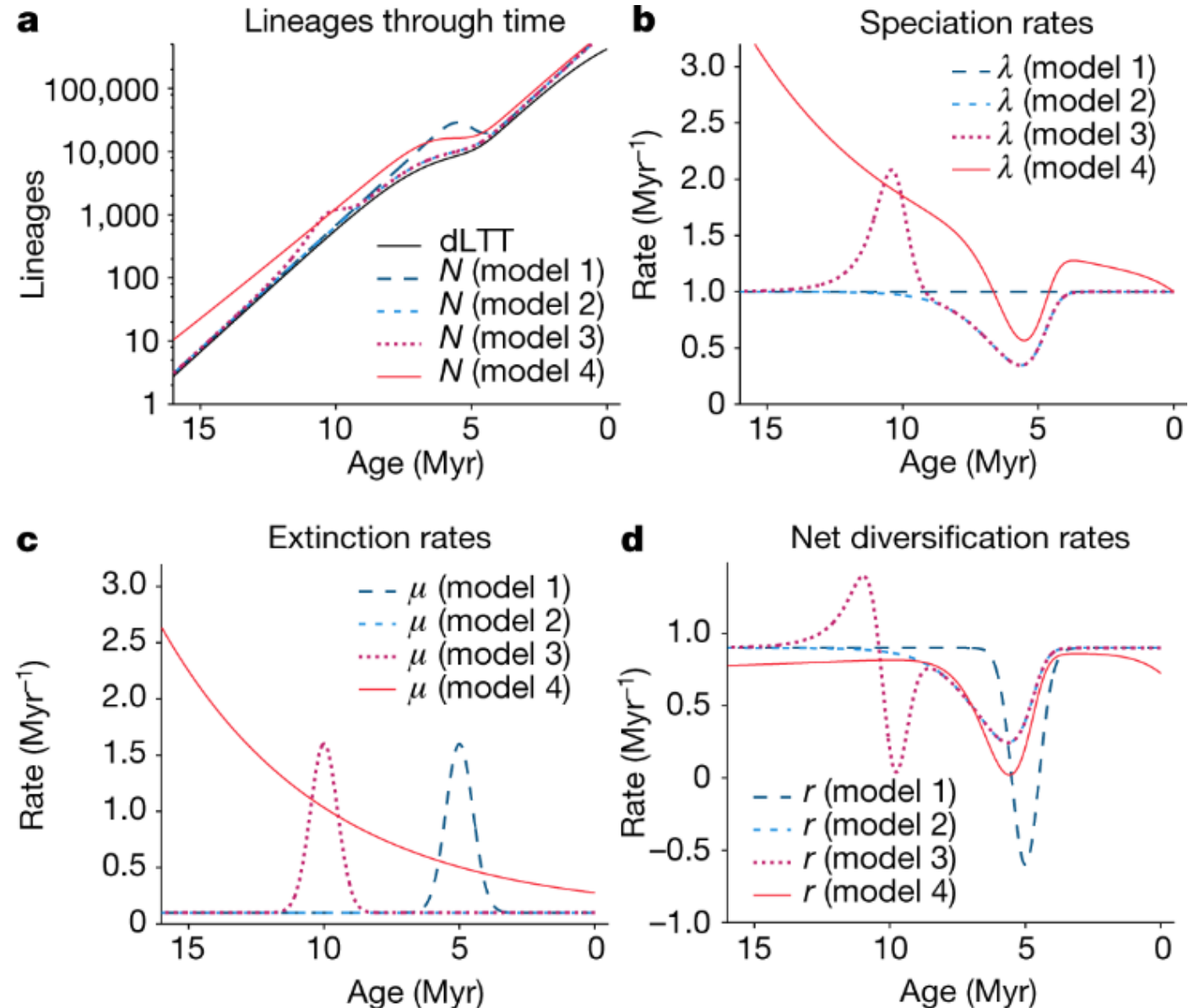
1. if the diversification rates varied between 0 and 1, but not between hidden states *a* and *b*, we could conclude that shifts in mating system explained all diversification rate heterogeneity
2. if the diversification rates did not vary between 0 and 1, but did vary between hidden states *a* and *b* we could conclude that there were background rate changes unassociated with mating system and that mating system evolution was not associated with rate shifts; or
3. if the diversification rates varied both between 0/1 and between hidden states *a/b*, then depending on the phylogenetic pattern of the hidden states they could represent the different long and short term consequences of the loss of SI

HiSSE disminuye los “falsos positivos” de BiSSE



Non-identifiability in BDP

- BiSSE y otros modelos basados en BDP incluyen más información que puede restringir el espacio de parámetros viables



Preguntas?

Simulaciones